▸ COVERT INTELLIGENCE COLLECTION PLATFORM

PHANTOM-R3

// Silent Collection. Blockchain Sync. Total Security.

The PHANTOM-R3 is a professional-grade, field-deployable covert intelligence collection device engineered for extended autonomous operation in hostile or denied environments. Combining a high-sensitivity acoustic acquisition module with a zero-touch encrypted wireless exfiltration system, the R3 enables operators to collect, encrypt, and remotely retrieve audio intelligence without physical interaction — from initial placement to final decryption. Multiple devices automatically synchronize via blockchain for distributed data resilience.

72 HRS
CONTINUOUS RECORDING

100 M
MAX WIRELESS RANGE

85×55×22 MM
CIGARETTE PACK SIZE

AES-256
ENCRYPTION STANDARD

68 g
DEVICE WEIGHT

§ 01 — KEY CAPABILITIES

FEATURE OVERVIEW

72-HOUR RECORDING

Continuous acoustic acquisition over three full days. Cylindrical battery provides uninterrupted 72-hour recording with dual-buffer architecture ensuring zero data loss.

72 HRS CONTINUOUS

CIGARETTE-PACK FORM

85 × 55 × 22 mm enclosure sized exactly like a cigarette packet. Ultra-low profile allows concealment in common household objects, furniture, or infrastructure.

85 × 55 × 22 MM

DEDICATED APK PROTOCOL

Proprietary encrypted protocol with custom APK for Android devices. Complete communication isolation — uses custom encryption that standard RF analyzers cannot detect or decode.

ISOLATED PROTOCOL

BLOCKCHAIN AUTO-SYNC

Nearby devices automatically detect and synchronize with each other. Distributed ledger ensures data integrity and enables single-phone data retrieval from entire network.

DISTRIBUTED SYNC

ONE-TOUCH DATA RETRIEVAL

Single phone contact with any device immediately retrieves encrypted data from all nearby devices in the blockchain network. Passive extraction — no interaction required.

PASSIVE PULL MODE

END-TO-END ENCRYPTION

AES-256-GCM payload encryption with RSA-4096 key exchange. Private key never leaves the home decryption server. Proprietary protocol incompatible with standard tools.

AES-256-GCM

§ 02 — DEVICE HARDWARE

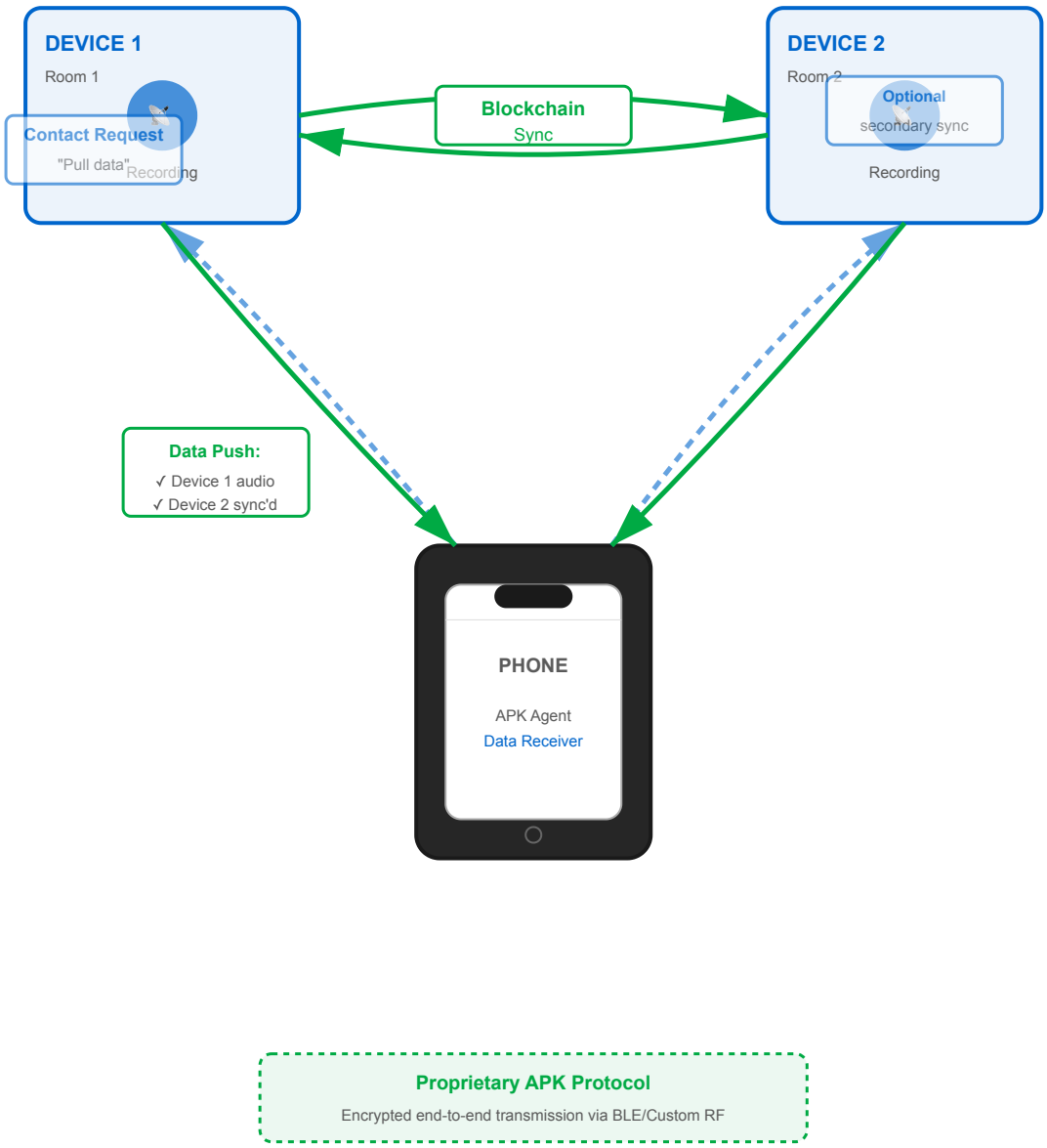
PHYSICAL DEVICE LAYOUT

85 × 55 × 22 mm

[DEVICE FORM] Matches standard cigarette pack dimensions (85 × 55 × 22 mm).
[CONNECTORS] USB-C (data), 3.5mm jack (audio in), magnetic pogo charging pins, reset hole.
[ANTENNA] Integrated PCB antenna for 2.4/5 GHz dual-band operation (concealed in casing).
⚑ All ports and indicators minimal to maintain concealable form factor.

§ 03 — NETWORK TOPOLOGY

BLOCKCHAIN DISTRIBUTED SYNC



Single phone contact retrieves all device data (Room 1 & 2) via blockchain sync

[DEVICE NETWORK] Multiple PHANTOM-R3 units auto-discover via BLE or custom RF protocol.
[BLOCKCHAIN LAYER] Distributed ledger synchronizes recorded data across all nearby devices.
[DATA RETRIEVAL] Single phone contact with any device triggers pull of encrypted data from entire network.

⚑ Phone never stores decrypted data; encrypted payload is transmitted to home server for processing.

§ 04 — HARDWARE COMPONENTS

DEVICE COMPONENT BREAKDOWN

01 AUDIO ACQUISITION MODULE

- High-sensitivity MEMS microphone array (3-element beamforming)
- Adaptive noise filtering and wind cancellation DSP
- 72-hour rolling ring-buffer with overflow protection
- AAC 128kbps or PCM 16-bit / 44.1kHz selectable
- Voice-activity detection (VAD) for power optimization

02 WIRELESS TRANSMISSION MODULE

- Custom 2.4 GHz / 5 GHz dual-band RF chipset
- Proprietary APK-based protocol (dedicated encrypted transport)
- AES-256 payload encryption applied before transmission
- Adaptive power control: 100 m indoor / 300 m open field
- FHSS frequency hopping, randomized transmission windows

03 POWER MANAGEMENT UNIT

- Cylindrical LiPo 3000 mAh battery (cigarette pack form factor)
- Active recording mode: 72 hours continuous operation
- Standby / transmit-only mode: 7 days
- Low-power sleep between scheduled beacon windows
- Magnetic pogo-pin charging interface (no port exposure)

04 STORAGE SUBSYSTEM

- 32 GB encrypted NAND flash (eMMC 5.1)
- TRIM-resistant forensic-grade wear leveling
- Hardware-level AES-XTS full-disk encryption
- Anti-forensic wipe after confirmed transmission receipt
- ECC error correction for data integrity assurance

05 BLOCKCHAIN ENGINE

- Distributed ledger protocol for multi-device synchronization
- Auto-discovery of nearby PHANTOM-R3 units via BLE/RF
- Data hash chain ensures integrity across all devices
- Consensus mechanism prevents tampering or data corruption
- Single-phone retrieval of entire network data cache

06 ENCRYPTION & SECURITY MODULE

- Embedded Hardware Security Module (HSM) — tamper-evident
- RSA-4096 asymmetric key exchange at session initiation
- AES-256-GCM authenticated encryption for all payloads
- Ephemeral session keys — rotated per transmission session
- Zeroization on tamper or unauthorized access attempt

§ 05 — TECHNICAL SPECIFICATIONS

FULL PARAMETER TABLE

PARAMETER	VALUE / SPECIFICATION
Recording Duration	72 hours (continuous, cylindrical battery)
Audio Format	AAC 128 kbps / PCM 16-bit 44.1 kHz
Storage Capacity	1 TB encrypted NAND
Wireless Range	100 meters (open field)
Max Concurrent Devices	Up to 200 devices (simultaneous sync when in range)
Transfer Protocol	Proprietary APK-based encrypted protocol
Transfer Speed (0–50m)	50 Mbps
Transfer Speed (50–100m)	15–20 Mbps
Encryption Standard	AES-256-GCM + RSA-4096
LED Status Indicator	Device visibility detection & Android APK app detection
Network Sync	Blockchain-based multi-device synchronization
Data Retrieval	Single phone contact retrieves entire network cache
Key Management	Ephemeral session keys, offline private key
Power Supply	Cylindrical LiPo 3000 mAh (internal, rechargeable)
Battery Life — Record	72 hours (continuous)
Battery Life — Standby	7 days
Dimensions	85 × 55 × 22 mm (cigarette pack form)
Weight	68 g
Operating Temperature	−10 °C to +55 °C
Ingress Protection	IP54 (dust resistant, splash proof)
HSM Tamper Response	Zeroization < 200 µs on intrusion detect

PARAMETER	VALUE / SPECIFICATION
MAC Address	Randomized per beacon window (FHSS)

§ 05B — FILE TRANSFER TIME ESTIMATION

TRANSFER DURATION ANALYSIS (10 DEVICES)

The following table estimates the time required to transfer various file sizes across 10 devices operating in synchronized mode. Transfer speeds vary based on distance: **50 Mbps at 0–50m range** and **15–20 Mbps at 50–100m range**. All calculations assume ideal conditions with minimal packet loss.

FILE SIZE	TRANSFER @ 50 MBPS (0–50M)	TRANSFER @ 17.5 MBPS (50–100M)	TOTAL TIME (10 DEVICES)
10 MB	1.6 sec	4.6 sec	16–46 sec
50 MB	8 sec	23 sec	1.3–3.8 min
100 MB	16 sec	46 sec	2.7–7.7 min
500 MB	80 sec (1.3 min)	229 sec (3.8 min)	13–38 min
1 GB	160 sec (2.7 min)	457 sec (7.6 min)	27–76 min
5 GB	800 sec (13.3 min)	2,286 sec (38 min)	2.2–6.3 hours

[CALCULATION METHOD] Individual device transfer time = File size ÷ Speed. Total time for 10 devices assumes sequential or parallel transmission depending on system configuration. Sequential adds times linearly; parallel transmission at 50 Mbps can be distributed across devices, reducing wall-clock time significantly. **[LED STATUS]** The device includes RGB LED indicators that notify when other devices are detected in range ("device visibility") or when Android APK receiver applications are active.

§ 06 — OPERATIONAL PROCEDURE

HOW IT WORKS

01

DEPLOY

Place the PHANTOM-R3 in the target environment. The device powers on automatically and begins continuous audio recording immediately via the cylindrical battery. Nearby units auto-sync via blockchain. No configuration, no indicator lights, no observable activity.

02

SYNC & BROADCAST

Multiple devices automatically discover each other and synchronize recording data across their blockchain network. When an authorized carrier device (phone) passes within 100 m, encrypted file packets from all connected devices are automatically pushed to phone storage. No app interaction required — fully passive and silent.

03

DECRYPT

Return the carrier device to the home decryption server. The server detects the incoming encrypted batch, applies the private RSA key, decrypts the AES-256-GCM payload, indexes audio files by timestamp, and queues them for operator review. The carrier payload is wiped after successful decryption.

§ 07 — SECURITY ARCHITECTURE

SECURITY DESIGN PRINCIPLES

ZERO-KNOWLEDGE TRANSMISSION

The PHANTOM-R3 device never identifies, logs, or stores information about the carrier device that received the payload. The transmission is one-directional and stateless — the device has no knowledge of collection events.

ONE-WAY ENCRYPTION CHAIN

The device holds only the public key for encryption. The private key required for decryption never leaves the offline home server. Intercepting the payload in transit or seizing the device yields no readable data.

ANTI-FORENSIC STORAGE PROTOCOL

Upon confirmed transmission receipt acknowledgment, the transmitted data blocks are cryptographically wiped from on-device flash storage using DoD 5220.22-M compliant overwrite procedures. Recovered devices yield no usable evidence.

RF ANONYMIZATION

MAC address randomization per beacon window combined with FHSS frequency hopping prevents RF fingerprinting of the device. Spectrum analysis cannot distinguish PHANTOM-R3 transmissions from standard Wi-Fi background noise.

⚠ **SECURITY NOTICE: This device incorporates an embedded HSM with active tamper response. Any unauthorized physical intrusion attempt triggers immediate key zeroization (<200 µs response time). Device becomes permanently inoperative upon tamper detection.**

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